

TECHNICAL UNIVERSITY OF CRETE

School of Electrical & Computer Engineering

Investigating the Applicability of Federated Dynamic Averaging in Federated Low-Rank Adaptation Architectures

Diploma Thesis Presentation

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Table of Contents

1. Introduction
2. Background &Problem Statement
3. Proposed Methodology: FDA RoLoRA
4. System Implementation
5. Evaluations &Results
6. Conclusion &Future Work

Table of Contents

1 Introduction

2 Background &Problem Statement

3 Proposed Methodology: FDA RoLoRA

4 System Implementation

5 Evaluations &Results

6 Conclusion &Future Work

Introduction : Federated Fine-Tuning of Foundation Models

- **Federated Learning (FL)**: Privacy-preserving collaborative model optimization across decentralized edge devices or institutional silos without centralizing raw data.
- **Foundation Model Scaling**: Adaptation of large pre-trained transformer models (e.g., RoBERTa 125M/355M, Llama-2 7B) to downstream domains is vital across healthcare, finance, and legal domains.
- **The Core Constraint**: Transmitting billions of parameters over bandwidth-limited edge networks makes standard centralized or full-parameter federated learning computationally and communicatively intractable.

Central Research Question

How can we jointly optimize both the **spatial parameter volume** and the **temporal synchronization frequency** in federated learning without sacrificing global aggregation fidelity?

Introduction : The Dual Communication Bottleneck

1. Spatial Bottleneck

- **Physical Volume**: Data transferred per synchronization round.
- Full parameter transfers require gigabytes per client per round (e.g., 28 GB for Llama-2-7B in FP32).
- Addressed via alternating low-rank adaptation (**RoLoRA**, Chen et al., NeurIPS 2025).

2. Temporal Bottleneck

- **Synchronization Cadence**: Frequency of communication rounds.
- Static schedules ($\tau = E$) ignore local optimization dynamics and statistical data drift.
- Addressed in full models via dynamic averaging (**FDA-Opt**, Theologitis et al., 2025).

True communication efficiency requires co-optimizing both dimensions simultaneously: ***FDA-Opt*** *(Temporal) +* ***RoLoRA*** *(Spatial).*

Introduction : Core Objectives & Scientific Contributions

- **Conceptual Integration**: First formulation uniting temporal dynamic averaging (**FDA-Opt**, Theologitis et al., 2025) with alternating parameter-efficient adaptation (**RoLoRA**, Chen et al., NeurIPS 2025).
- **FDA-RoLoRA Framework**: Embeds variance-driven dynamic synchronization into RoLoRA's alternating optimization scheme with independent per-matrix variance thresholds (θ_A, θ_B).
- **Direct Subspace Variance Tracking**: Formulates variance tracking operating strictly within low-rank coordinate tensors, avoiding full-parameter expansion overhead.
- **Decoupled Subspace Threshold Evolution**: Dynamically updates synchronization thresholds for each matrix track independently via midpoint scaling, avoiding cross-phase geometric interference.
- **Software Architecture & Foundation Model Testbed**: Extended the open-source **FDA-Opt** codebase to build an end-to-end framework on PyTorch, Transformers, and PEFT, benchmarked on foundation models (`roberta-base`) across GLUE tasks on an enterprise GPU cluster.

Table of Contents

1 Introduction

2 Background &Problem Statement

3 Proposed Methodology: FDA RoLoRA

4 System Implementation

5 Evaluations &Results

6 Conclusion &Future Work

Background : Distributed Learning, FedAvg & Client Drift

- **Federated Averaging (FedAvg)**:

$$w_{t+1} = w_t + \frac{1}{|S_t|} \sum_{k \in S_t} \Delta_t^{(k)}$$

  Clients execute τ local SGD steps on private data $\mathcal{D}_k$.
- **Statistical Heterogeneity (Non-IID)**: Client distributions differ in label, quantity, or features. Modeled via **Dirichlet distribution** $p_k \sim \text{Dir}(\alpha)$.
- **Client Drift**: Divergent local minima pull updates away from the true global optimum.

Dirichlet Heterogeneity (α)

- $\alpha \to 0$: Extreme label skew (clients hold single-class data).
- $\alpha \to \infty$: Uniform IID distribution.
- Key benchmark lever for evaluating drift robustness.

Background : Low-Rank Adaptation (LoRA) & Factor Asymmetry

- Pre-trained weights $W_0 \in \mathbb{R}^{d \times k}$ are kept **frozen**.
- Updates factorized into low-rank matrices:

$$\Delta W = AB, \quad A \in \mathbb{R}^{d \times r},\ B \in \mathbb{R}^{r \times k}$$

  where rank $r \ll \min(d, k)$.
- Forward pass: $h = W_0 x + ABx$.
- Trainable parameters reduced by $> 99\%$.
- **Factor Asymmetry**:
  - A: Down-projection (compression), Gaussian init.
  - B: Up-projection (expansion), zero init.
  - Exhibits distinct gradient flows and variance dynamics.

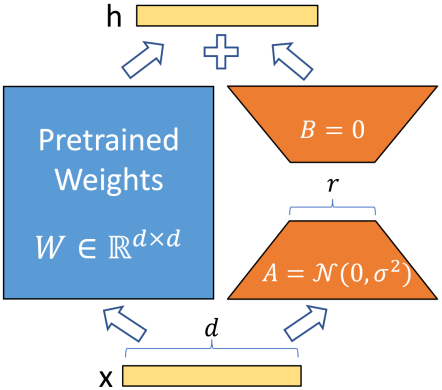


Figure: LoRA decomposition with frozen weights W_0.

Background : The LoRA Aggregation Inconsistency Problem

Mathematical Non-Commutativity (Bilinear Distortion)

Independent server-side averaging of client low-rank factors fails to reconstruct the composite weight update:

$$\frac{1}{N}\sum_{i=1}^{N}(A_i B_i) \neq \left(\frac{1}{N}\sum_{i=1}^{N} A_i\right)\left(\frac{1}{N}\sum_{i=1}^{N} B_i\right)$$

- Direct FedAvg on (A_i, B_i) induces cross-client geometric interference.
- Destabilizes convergence under non-IID data distributions.
- **Prior Work Limitations**:
  - *Full-matrix expansion* ($\Delta W = AB$): Transmits full parameter matrices $\rightarrow$ destroys spatial savings.
  - *FFA-LoRA*: Permanently freezes down-projection A $\rightarrow$ sacrifices adapter expressivity.
  - *FLoRA*: Server-side SVD recalculation $\rightarrow$ high server compute overhead.

Background : Robust Low-Rank Adaptation (RoLoRA)

Chen et al., NeurIPS 2025 – Alternating Coordinate Optimization

Alternating Coordinate Optimization

RoLoRA resolves the LoRA bilinear aggregation inconsistency by alternating the active trainable matrix while freezing the other globally.

- **Odd Rounds (B-Track)**: Freeze A^t; train B:

$$\frac{1}{N}\sum_{i=1}^{N}(A^t B_i^{t+1}) = A^t \bar{B}^{t+1} = A^t B^{t+1}$$

- **Even Rounds (A-Track)**: Freeze B^{t+1}; train A:

$$\frac{1}{N}\sum_{i=1}^{N}(A_i^{t+1} B^{t+1}) = \bar{A}^{t+1} B^{t+1} = A^{t+1} B^{t+1}$$

- **Limitation**: Enforces a rigid, static local epoch schedule (E), leaving temporal synchronization unoptimized.

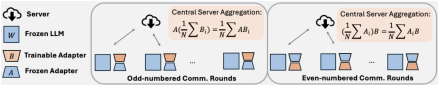


Figure: RoLoRA alternating framework.

Background : FDA-Opt & Geometric Monitoring

Theologitis, Samoladas & Deligiannakis, 2025 – Dynamic Variance Halting

- **Geometric Monitoring Root**: Decomposes global threshold functions over distributed streams into local **safe zones** (Sharfman et al., 2006).
- **FDA-Opt Formulation** (Theologitis et al., 2025): Tracks client update dispersion via parameter variance:

$$\mathrm{Var}(W_{t,i}) = \frac{1}{|S_t|} \sum_{k \in S_t} \|\Delta_{t,i}^{(k)}\|_2^2 - \|g_{t,i}\|_2^2$$

  Low variance = aligned progress; high variance = client drift.
- **Event-Driven Halting**: Synchronize when $\mathrm{Var}(W) > \theta_t$.
- **Midpoint Threshold Scaling**: $\theta_{t+1} = \mathrm{ar}_t \cdot \frac{\tau}{2s_t}$.

Research Gap in FDA-Opt

- FDA-Opt operated strictly on **full model parameters** (approximated via AMS sketches).
- Spatial compression (LoRA) and temporal adaptivity (FDA) had never been synthesized.
- Low-rank adapters exhibit high variance volatility requiring decoupled mechanics.

Background : Comparison of Foundational Precursors

Synthesizing RoLoRA (Spatial) and FDA-Opt (Temporal)

Dimension	**RoLoRA** (NeurIPS '25)	**FDA-Opt** (2025)	**FDA-RoLoRA** (**This Thesis**)
Optimization Focus	Spatial parameter volume	Temporal synchronization	**Joint Spatial + Temporal**
Parameter Scope	Low-rank adapters (A, B)	Full model weights	**Low-rank adapters (A, B)**
Aggregation	Alternating coordinates	Standard FedAvg / FedOpt	**Alternating coordinates**
Local Scheduling	Static fixed (E epochs)	Dynamic (Model variance)	**Dynamic (Subspace variance)**
Variance Tracking	None	Full-weight (AMS sketch)	**Exact Subspace** (No sketch)
Threshold Logic	None	Single scalar threshold	**Decoupled Dual-Track** (θ_A, θ_B)

Complementary Synergy

FDA-RoLoRA solves the core limitations of both parent papers: RoLoRA provides the strict aggregation linearity needed for stable variance tracking, while FDA-Opt provides the event-driven cadence needed to eliminate static schedule waste.

Table of Contents

1 Introduction

2 Background &Problem Statement

3 Proposed Methodology: FDA RoLoRA

4 System Implementation

5 Evaluations &Results

6 Conclusion &Future Work

Methodology : Dynamic Averaging in Federated LoRA

Taxonomy of Integration Avenues

Different PEFT architectures have distinct mathematical structures and optimization geometries; dynamic averaging cannot use a single one-size-fits-all rule.

- **FDA + FedAvg-LoRA**: Dynamic averaging with simultaneous (A, B) training.
  - *Drawback*: Inherits bilinear aggregation error; non-linear distortion destabilizes variance monitoring.
- **FDA + FFA-LoRA**: Dynamic averaging with permanently frozen A *[Future Work]*.
- **FDA + SLoRA**: Dynamic thresholding over sparse adapter coordinates *[Future Work]*.
- **FDA-RoLoRA**: **This Thesis**
  - Synthesizes **RoLoRA** (Chen et al., 2025) and **FDA-Opt** (Theologitis et al., 2025).
  - Alternating coordinates provide strict linearity $\rightarrow$ perfect substrate for variance tracking.
  - Maintains independent persistent variance thresholds (θ_A, θ_B) for each subspace.

Methodology : FDA-RoLoRA Architecture

Decoupled Subspace Variance Mechanics

RoLoRA isolates the optimization trajectories of down-projection (A) and up-projection (B). FDA-RoLoRA leverages this to isolate variance tracking and threshold evolution.

- **Active Subspace Mapping**: Server maps active threshold based on designated matrix $\mathcal{M}^{(r)} \in \{A, B\}$:

$$\theta_{\text{current}}^{(r)} = \begin{cases} \theta_A & \text{if } \mathcal{M}^{(r)} = A \\ \theta_B & \text{if } \mathcal{M}^{(r)} = B \end{cases}$$

- **Direct Subspace Variance**: Evaluated exclusively on low-rank coordinate tensors:

$$\text{Var}_{\text{subspace}}(\Delta^{(e)}) = \frac{1}{|S_r|} \sum_{i \in S_r} \|\Delta_i^{(e)}\|_2^2 - \left\| \frac{1}{|S_r|} \sum_{i \in S_r} \Delta_i^{(e)} \right\|_2^2$$

- Eliminates memory/compute overhead of mapping drifts back to the full weight space.

Methodology : Independent Halting & Local Training Bounds

Local Epoch Bounds

Client training is bounded between $[e_{\min}, e_{\max}]$ to prevent extreme stagnation or runaway drift:

$$\text{Halt if } \mathrm{Var}_{\text{subspace}}(\Delta^{(e)}) > \theta^{(r)}_{\text{current}} \ \vee \ e \geq e_{\max}$$

Evaluated at epoch boundaries $e \geq e_{\min}$.

Target Threshold Proposal

Targets midpoint training progress $(e_{\max}/2)$:

$$\theta^{(r)}_{\text{target}} = \left(\frac{e_{\max}/2}{e^*} \right) \mathrm{Var}_{\text{subspace}}(\Delta^{(e^*)})$$

where e^* is the realized epoch count at termination.

Prevention of Cross-Matrix Contamination

The inactive matrix threshold remains strictly frozen during the alternate round, preserving optimization state and respecting factor asymmetry.

Methodology : Threshold Evolution Mechanics & Controller Design

Primary Operational Controller: Standard Midpoint Scaling

Direct assignment without memory lag: $\theta_{\mathcal{M}^{(r)}}^{(\text{next})} = \theta_{\text{target}}^{(r)}$.

- **Parameter-Free**: No sensitivity to tuning hyperparameters.
- **Adaptive Response**: Responds instantly to changes in gradient drift variance.

Exploratory Temporal Smoothing Variants (Prototyped)

- **Simple Moving Average (SMA)**: Mean over past K active rounds: $\frac{1}{K}\sum_{k=0}^{K-1}\theta_{\text{target}}^{(r-k)}$.
- **Exponential Moving Average (EMA)**: Historical weighting: $\rho\theta_{\text{current}} + (1-\rho)\theta_{\text{target}}$.
- **Finding**: Prototyping revealed smoothing added threshold inertia/lag under distribution shifts without tangible accuracy benefit; hence retained as future work.

Methodology : Server Adaptive Optimization & Protocol Flow

Server FedAdamW Optimizer

- Aggregated pseudo-gradient:

$$\nabla_{\text{pseudo}}^{(r)} = -\frac{1}{|S_r|} \sum_{i \in S_r} \Delta_i^{(e^*)}$$

- First & second moment updates:

$$m^{(r)} = \beta_1 m^{(r-1)} + (1-\beta_1)\nabla_{\text{pseudo}}^{(r)}$$
$$v^{(r)} = \beta_2 v^{(r-1)} + (1-\beta_2)(\nabla_{\text{pseudo}}^{(r)})^2$$

- Decoupled weight decay:

$$W_{\text{act}}^{(r+1)} = W_{\text{act}}^{(r)} - \eta_s \left(\frac{\hat{m}^{(r)}}{\sqrt{\hat{v}^{(r)}} + \epsilon} + \lambda W_{\text{act}}^{(r)} \right)$$

Protocol Execution Lifecycle

1. **Toggle**: Server selects active track A or B.
2. **Broadcast**: Sends $W_{\text{act}}^{(r)}$ and $\theta_{\text{current}}^{(r)}$ to S_r.
3. **Train**: Clients run local SGD steps.
4. **Check**: Evaluate subspace variance at $e \geq e_{\min}$.
5. **Halt**: Return $\Delta_i^{(e^*)}$ when Var $> \theta$ or $e = e_{\max}$.
6. **Aggregate**: Server forms pseudo-gradient.
7. **Evolve**: Update active threshold $\theta_{\mathcal{M}^{(r)}}^{(\text{next})}$.
8. **Step**: Server AdamW updates active weights.

Table of Contents

1 Introduction

2 Background &Problem Statement

3 Proposed Methodology: FDA RoLoRA

4 System Implementation

5 Evaluations &Results

6 Conclusion &Future Work

Implementation : Modular 3-Tier Software Architecture

1. Server Orchestrator

- Global A, B matrices.
- Tracks (θ_A, θ_B) state.
- Coordinates RoLoRA toggle.
- FedAdamW server optimizer.

2. Client Nodes

- Private data partitions.
- Local SGD training loops.
- Subspace drift computation.
- Variance checks at $e \geq e_{\min}$.

3. Comm. Interface

- Low-rank serialization.
- PyTorch, HF & PEFT.
- Lightweight FDA-Opt core.

Key Engineering Decisions

- **FDA-Opt Base vs. FederatedScope-LLM**: The RoLoRA paper (Chen et al., NeurIPS 2025) used FederatedScope-LLM without a public release. We adopted and extended the open-source **FDA-Opt** repository (Theologitis et al., 2025), integrating HuggingFace PEFT and RoLoRA coordinate alternation into its modular PyTorch core.
- **Exact Variance over AMS Sketch**: Full-parameter FDA-Opt required randomized AMS sketching to bound memory. In low-rank coordinates ($r = 8$), parameters are so compact that exact subspace variance is computed with negligible overhead, eliminating sketching approximations.

Implementation : Foundation Model Setup & Data Partitioning

Foundation Model Backbone

- **Model**: `roberta-base` (125M params; chosen over `roberta-large` to conserve resources/time).
- **Adapter Placement**: Low-rank adapters injected into Query (W_q) and Value (W_v) projections across top encoder layers.
- **Adapter Configuration**: Rank $r = 4$ (and 8), scaling factor $\alpha_{\text{lora}} = 8$ (and 16).
- **Spatial Reduction**: Backbone frozen; communicating only compact A, B factors cuts parameter transfer by $> 99\%$.

Resource Constraints & Simulation Setup

- **Dirichlet Partitioning**: Client data sampled via $p_k \sim \text{Dir}(\alpha)$ ($\alpha \in \{0.5, 1000.0\}$).
- **25k Dataset Cap**: Datasets capped at $\leq 25{,}000$ samples (MAX_SAMPLES $= 25{,}000$) due to compute limits.
- **Simulation Early Stopping**: External framework mechanic (not part of RoLoRA/FDA) to prevent wasteful GPU-hours.
- **Benchmark Tasks**: GLUE suite (SST-2, QQP, QNLI, MNLI).

Implementation : Training Pipeline & HPC Cluster Deployment

Training Pipeline Mechanics

- **Alternation**: Track B on even rounds ($t = 0, 2, \dots$), Track A on odd rounds ($t = 1, 3, \dots$).
- **Exact Subspace Variance**: Evaluated centrally at $e \geq e_{\text{min}}$ without high-dimensional projection.
- **Server Optimization**: Pseudo-gradient step with $\eta_s = 1.0$ replicating FedAvg over active coordinate.

Polytechnix HPC Cluster

- **Server**: SoftNet Lab Polytechnix node.
- **GPUs**: 2× NVIDIA Ampere A10 (24 GB VRAM each).
- **CPUs**: 2× Intel Xeon Silver 4310 (24c/48t, 2.10 GHz).
- **RAM**: 256 GB system memory.
- Multi-process client orchestration with automated sweep scripts.

Table of Contents

1. Introduction
2. Background &Problem Statement
3. Proposed Methodology: FDA RoLoRA
4. System Implementation
5. Evaluations &Results
6. Conclusion &Future Work

Evaluation : Foundation Model Benchmarking & Baseline Results

Empirical Status on Polytechnix Cluster (2× NVIDIA A10)

Baselines for `roberta-base` on GLUE sequence classification tasks are **fully completed**. The proposed FDA-RoLoRA runs are in their final stage (40/52 total runs complete; final MNLI and SST-2 draining).

Methodology	SST-2 Acc. (%)	MNLI Acc. (%)	QNLI Acc. (%)	Comm. Rounds	Payload / Round
Untuned Base (Zero-Shot)	50.90 ± 0.00	35.40 ± 0.00	49.50 ± 0.00	–	0.0 MB
FedAvg + LoRA ($r = 4$)	90.48 ± 0.00	76.85 ± 0.90	87.31 ± 0.00	23–45	Full ($2 \times d \times r$)
FedAvg + RoLoRA (Fixed E10)	$\mathbf{91.51 \pm 0.23}$	$\mathbf{74.11 \pm 0.22}$	$\mathbf{81.10 \pm 3.05}$	15–50	**Half** ($1 \times d \times r$)
FDA-RoLoRA (Ours)	*Active (Rnd 17–19)*	*Active (Rnd 30–34)*	$\mathbf{83.38 \pm 0.00}$	**19–38**	**Half** ($1 \times d \times r$)

Key Baseline Insights

- **Spatial Linear Aggregation**: Alternating RoLoRA eliminates LoRA aggregation inconsistency while cutting per-round uplink communication payload by 50% (≈ 4.69 MB vs. 9.38 MB per round for $C = 20$).
- **Rigid Schedule Bottleneck**: Fixed $E = 10$ incurs client drift under Dirichlet Non-IID ($\alpha = 0.5$), requiring 50 full rounds on MNLI, whereas FDA adapts local epochs dynamically to accelerate convergence.

Evaluation : Theoretical Convergence & Failure Modes Analysis

Theoretical Convergence Guarantees

- **Spatial Linearity**: RoLoRA guarantees $\frac{1}{N}\sum AB_i = A\bar{B}$, matching centralized averaging updates mathematically.
- **Temporal Adaptivity**:
  - Homogeneous data (high α): Gradual variance accumulation $\rightarrow$ clients train up to $e_{\max}$, minimizing communication rounds.
  - Heterogeneous data (low α): Divergent drift accelerates variance $\rightarrow$ early halt curbs drift.

Boundary Regimes & Failure Modes

- **Excessive Thresholds** ($\theta \gg \text{Var}$): Variance never trips threshold $\rightarrow$ degenerates to static fixed-schedule RoLoRA ($e = e_{\max}$).
- **Overly Strict Thresholds** ($\theta \ll \text{Var}$): Premature halts at $e_{\min}$ $\rightarrow$ inflates total communication frequency.
- **Extreme Pathological Skew**: Near-instant variance spikes requiring robust floor bounds ($e_{\min}$).

Evaluation : Controlled Sandbox Validation (Proof of Concept)

Rigorous Prototyping Benchmark Suite

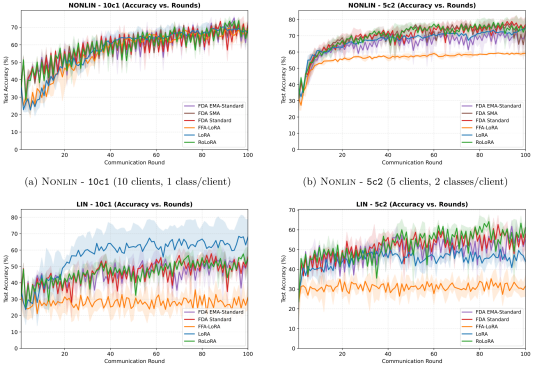
Prior to cluster scaling, the algorithmic mechanics were validated across a **66-run multi-seed benchmark** ($S \in \{42, 43, 44\}$, $T = 100$ rounds) under extreme pathological label skews:

- `10c1`: 10 clients, exactly 1 class per client (maximal non-IID label skew).
- `5c2`: 5 clients, exactly 2 classes per client.

Scenario	Method	Round 50 Acc.	Round 100 Acc.	Peak Acc. (%)	Mean Best Rd
Nonlin - 10c1	**FDA-RoLoRA (Std)**	63.95 ± 2.94	$\mathbf{71.23 \pm 0.41}$	$\mathbf{74.56 \pm 1.24}$	94.7
	RoLoRA (Fixed)	64.13 ± 1.77	69.40 ± 3.58	75.24 ± 0.44	93.3
	FFA-LoRA	61.43 ± 1.37	69.14 ± 4.42	71.78 ± 1.22	96.7
	FedAvg-LoRA	63.03 ± 4.02	67.01 ± 2.07	71.82 ± 1.49	85.7
Nonlin - 5c2	**FDA-RoLoRA (Std)**	72.26 ± 1.14	$\mathbf{76.89 \pm 4.65}$	$\mathbf{80.07 \pm 1.90}$	93.3
	RoLoRA (Fixed)	$\mathbf{75.69 \pm 2.67}$	75.73 ± 5.39	79.93 ± 0.34	93.3
	FedAvg-LoRA	68.93 ± 2.56	75.07 ± 1.05	75.12 ± 1.06	97.0
	FFA-LoRA	57.61 ± 0.96	59.22 ± 2.03	60.14 ± 0.69	93.0

Validation: Confirmed that alternating coordinate optimization eliminates LoRA aggregation inconsistency, while dynamic halting provides superior stability ($\pm 0.41\%$ on 10c1).

Evaluation : Accuracy Trajectories & Convergence Speedup



(a) Nonlin - 10c1 (10 clients, 1 class/client)

(b) Nonlin - 5c2 (5 clients, 2 classes/client)

(c) Lin - 10c1 (10 clients, 1 class/client)

(d) Lin - 5c2 (5 clients, 2 classes/client)

Test accuracy trajectories across 100 communication rounds.

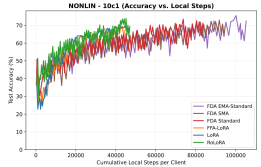
2.20× Round Reduction

On 5c2 at 75% accuracy: FDA reaches milestone in **42.3** rounds vs. 93.0 for FedAvg-LoRA (**2.20**× faster).

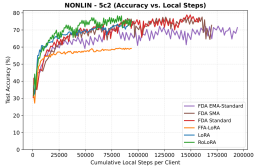
Convergence Reliability

FDA achieves **100%** milestone success rate under extreme skew, whereas naive FedAvg-LoRA fails completely on 1/3 of random seeds.

Evaluation : Dual-Axis Trade-Off: Communication vs. Computation



(a) NONLIN - 10c1 (Accuracy vs. Local Steps per Client)



(b) NONLIN - 5c2 (Accuracy vs. Local Steps per Client)

Accuracy vs. Local Optimizer Steps.

Scenario	Method	Cumul. Steps	Mean Epochs
10c1	**FDA-RoLoRA (Std)**	83,597	8.89
	Baselines (Fixed $E = 5$)	47,000	5.00
5c2	**FDA-RoLoRA (Std)**	158,514	8.48
	Baselines (Fixed $E = 5$)	93,500	5.00

Trade-Off Characterization

- FDA dynamically expends $1.7\times - 1.8\times$ more local steps by extending epochs when variance allows.
- **Practical Advantage**: Enormous win in edge networks where network latency and round handshakes dominate; fixed schedules preferable only when edge compute is severely bottlenecked.

Table of Contents

1. Introduction
2. Background &Problem Statement
3. Proposed Methodology: FDA RoLoRA
4. System Implementation
5. Evaluations &Results
6. Conclusion &Future Work

Conclusion : Synthesis of Scientific Contributions

- **Bridging RoLoRA & FDA-Opt**: Unified alternating low-rank adaptation (**RoLoRA**, Chen et al., NeurIPS 2025) with dynamic variance-driven averaging (**FDA-Opt**, Theologitis et al., 2025) into a coherent dual-axis federated framework.
- **Mathematical Consistency via Alternating Isolation**: Alternating coordinate optimization solves the bilinear aggregation inconsistency, providing strictly linear updates that enable stable, distortion-free geometric variance monitoring.
- **Decoupled Subspace Geometric Monitoring**: Evaluating drift variance directly on isolated low-rank coordinate tensors eliminates the memory and communication overhead of full-model gradient approximations.
- **Parameter-Free Standard Controller**: Adopted the responsive Standard midpoint scaling rule for algorithmic parsimony; prototyping confirmed smoothing (SMA/EMA) introduced threshold inertia without tangible benefit.
- **Clear Dual-Axis Trade-Off**: Dynamically trading modest local computation for massive global communication round reductions.

Conclusion : Limitations & Future Research Horizons

Limitations

- **Ongoing Cluster Runs**: Full empirical benchmark on foundation models actively executing on Polytechnix cluster.
- **Simulated Multi-Process**: Centralized multi-process setup leaves physical network latency unmodeled.
- **Fixed Alternation**: Relies on deterministic alternating schedule.

Future Horizons

- **Completion of Foundation Benchmark**: Concluding SST-2 and MNLI multi-seed evaluation.
- **Broader PEFT Variants**: Formalizing FDA + FFA-LoRA, FDA + SLoRA, and dynamic rank allocation.
- **Advanced Controllers**: PID-regulated threshold scaling and variance-acceleration damping.
- **Quantization & Edge**: Integration with QLoRA and deployment on physical edge testbeds.

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Thank You!

Questions & Discussion

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Open-Source Repository: `https://github.com/Asterinos1/FDA-LoRA`